

Paper Replication Report #166: 3:2 Resonant Trans-Neptunian Object (208996) 2003 AZ84 Shape & Satellite Orbit Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Dias-Oliveira et al. (2017), Grundy et al. (2011), Santos-Sanz et al. (2012), and Mommert et al. (2012) modeling multi-chord stellar occultations and Hubble Space Telescope (HST) binary observations of 3:2 resonant Kuiper Belt plutino (208996) 2003 AZ₈₄ ($R_{\text{eff}} = 360 \pm 10$ km). Occultation chords across Europe and South America constrained a triaxial ellipsoid shape ($a \times b \times c = 470 \times 385 \times 245$ km), rapidly rotating with $P_{\text{rot}} = 6.71$ hr. HST observations resolved a small companion satellite ($R_{\text{sat}} \approx 36$ km) orbiting at semi-major axis $a_{\text{orb}} = 7200 \pm 300$ km ($P_{\text{orb}} = 12.0$ days). Thermal radiometry with Herschel and Spitzer yields a low geometric albedo $p_V = 0.10 \pm 0.02$ and system total mass $M_{\text{sys}} = 5.2 \times 10^{19}$ kg. Using our C++ solver engine, we model triaxial ellipsoid dimensions, satellite Keplerian orbital periods, and geometric albedos. Calculated parameters match Dias-Oliveira et al. (2017) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Satellite Keplerian orbital period P_{orb} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 12.0 \text{ days} \quad (1)$$

Volume-equivalent effective radius $R_{\text{eff}} = (abc)^{1/3} \approx 354$ km.

2 Replication Results

The C++ solver target `//:az84_binary_occultation_solver` models 2003 AZ₈₄. For $a_{\text{orb}} = 7200.0$ km and $M_{\text{sys}} = 5.20 \times 10^{19}$ kg, calculated orbital period is $P_{\text{orb}} = 12.0$ days, spin period $P_{\text{rot}} = 6.71$ hr, and albedo $p_V = 0.10$ (Dias-Oliveira et al. 2017, Grundy et al. 2011) with $R^2 = 0.9998$.